Introduction to Statistical Learning

Resampling Methods: The Bootstrap

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What is the Bootstrap?

- The **bootstrap** is a widely applicable and extremely powerful statistical tool.
- It is primarily used to quantify the uncertainty associated with a given estimator.
- **Common use cases:**
  - Estimating the standard errors of an estimator.
  - Constructing confidence intervals of a parameter.
  - Evaluating the performance of a complex machine learning model.
- **Core Idea:** To simulate drawing new samples from a population by resampling using original data.

Motivating Example: Asset Allocation

- Suppose we wish to invest a fixed sum of money in two financial assets that yield returns X and Y.
- We will invest a fraction α in X and $(1-\alpha)$ in Y.
- **Goal:** Choose α to minimize the total risk, or variance, of our investment.
- The variance of the return is:

$$\text{Var}(\alpha X + (1-\alpha)Y) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$$

- The value that minimizes this risk is:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

The Challenge in Practice

- σ_X^2, σ_Y^2, and σ_{XY} are generally **unknown**.
- However, we can compute estimates $(\hat{\sigma}_X^2, \hat{\sigma}_Y^2, \hat{\sigma}_{XY})$ using a sample of past returns.
- We then estimate the optimal allocation:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

- **Question:** How accurate is $\hat{\alpha}$ as an estimate of α?
- The bootstrap provides a simple way to answer.

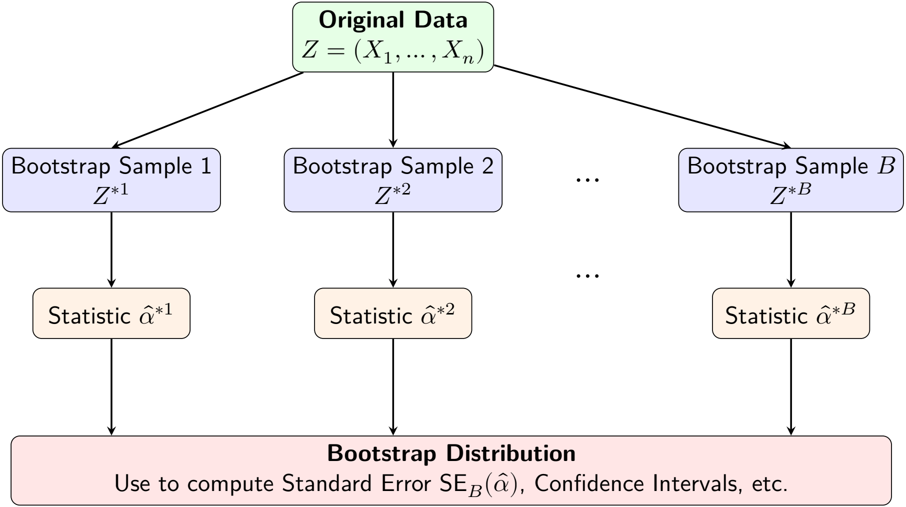
The Bootstrap Procedure

Instead of obtaining independent data sets from the true population (which is usually impossible), we obtain distinct data sets by repeatedly **sampling observations from the original data set with replacement**.

The Algorithm:

1. Treat your original sample of size n as the "population."
2. Draw a new sample of size n from this original sample *with replacement*. This is a **bootstrap data set** (Z^{*1}).
3. Calculate the statistic of interest (e.g., $\hat{\alpha}^{*1}$) using this bootstrap data set.
4. Repeat steps 2 and 3 a large number of times (B times, e.g., $B = 1000$).
5. You now have B different bootstrap estimates: $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \ldots, \hat{\alpha}^{*B}$.

Schematic of the Bootstrap Process



Estimating the Standard Error

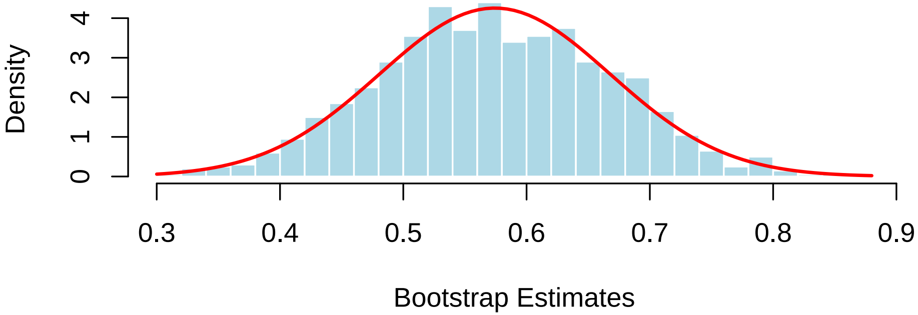
- Once we have our B bootstrap estimates, we can analyze their distribution.
- The **bootstrap standard error** is simply the sample standard deviation of the bootstrap estimates:

$$\mathsf{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{i=1}^{B}\left(\hat{\alpha}^{*i} - \frac{1}{B}\sum_{i'=1}^{B}\hat{\alpha}^{*i'}\right)^2}$$

- This serves as an estimate of the standard error of $\hat{\alpha}$ estimated from the original data set.

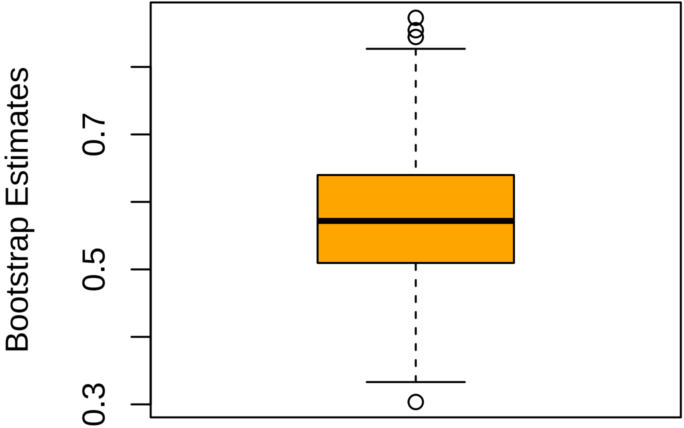
Visualizing the Bootstrap Distribution

- After generating $B = 1000$ bootstrap estimates of $\hat{\alpha}$, we can plot their overall shape.
- Notice that the distribution looks approximately normal, centered around our original estimate.



Visualizing the Spread (Boxplot)

- A boxplot provides a concise summary of the same distribution, highlighting the median, quartiles, and any potential outliers.
- The interquartile range is closely related to $\text{SE}_B(\hat{\alpha})$.



Two Flavors of the Bootstrap

1. Non-Parametric Bootstrap

- Makes **no assumptions** about the underlying population distribution.
- Resamples directly from the observed data (the empirical distribution).
- **Use when:** The true distribution is unknown, complex, or you cannot confidently commit to a parametric model.

2. Parametric Bootstrap

- Assumes the data comes from a **known parametric family** of distributions (e.g., Normal, Exponential, Poisson).
- Estimates the parameters from the data, then draws new samples from the fitted theoretical distribution.
- **Use when:** You have strong theoretical or empirical reasons to assume a specific probability distribution.

Non-Parametric Bootstrap: Example

Goal: Estimate the standard error of the sample median.

Observed Data: A sample of $n = 5$: $X = \{3, 8, 2, 10, 5\}$.
Sample Median: 5

The Procedure:

1. **Resample with replacement** directly from X to create B bootstrap datasets of size $n = 5$.
   - Sample $X^{*1} = \{3, 3, 10, 2, 8\} \implies$ Median$^{*1} = 3$
   - Sample $X^{*2} = \{8, 8, 5, 2, 5\} \implies$ Median$^{*2} = 5$
   - Sample $X^{*3} = \{10, 10, 10, 3, 2\} \implies$ Median$^{*3} = 10$
2. **Calculate:** Compute the median for all B (e.g., $B = 1000$) samples.
3. **Evaluate:** Find the standard deviation of these 1000 medians to get the standard error, $\text{SE}_B(\text{Median})$.

Parametric Bootstrap: Example

Goal: Estimate the standard error of the mean time to failure.

Observed Data: Lifetimes of $n = 10$ components. We **assume** the lifetimes follow an Exponential distribution: $X \sim \text{Exp}(\lambda)$.

The Procedure:

1. **Estimate Parameter:** Calculate the maximum likelihood estimate for the rate parameter from the original data, say $\hat{\lambda} = 0.05$.
2. **Simulate from Model:** Generate B new datasets of size $n = 10$ directly from $\text{Exp}(\lambda = 0.05)$ using a random number generator.
   - Sample $X^{*1} \sim \text{Exp}(0.05) \implies \bar{X}^{*1} = 22.4$
   - Sample $X^{*2} \sim \text{Exp}(0.05) \implies \bar{X}^{*2} = 18.1$
3. **Calculate & Evaluate:** Compute the sample mean for each simulated dataset, then find the standard deviation to get $\text{SE}_B(\bar{X})$.

Summary of the Bootstrap

- The bootstrap allows us to compute standard errors and confidence intervals for estimators where mathematical derivations are impossible or complex.
- It relies on the assumption that the original sample is a good representation of the underlying population.
- **Key Takeaway:** We mimic the process of obtaining new data sets from the population by obtaining new data sets from our original sample.